$$\sigma_z \otimes I = \begin{pmatrix} \langle uu|uu\rangle & \langle uu|ud\rangle & \langle uu|du\rangle & \langle uu|dd\rangle \\ \langle ud|uu\rangle & \langle ud|ud\rangle & \langle ud|du\rangle & \langle ud|dd\rangle \\ -\langle du|uu\rangle & -\langle du|ud\rangle & -\langle du|du\rangle & -\langle du|dd\rangle \\ -\langle dd|uu\rangle & -\langle dd|ud\rangle & -\langle dd|du\rangle & -\langle dd|dd\rangle \end{pmatrix}. \tag{7.3}$$

Because these eigenvectors are orthonormal, the matrix reduces to

$$\sigma_z \otimes I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{7.4}$$

How do we write the eigenvectors |*uu*>, |*ud*>, |*du*>, and |*dd*> as column vectors? For now, I'll just tell you that we'll represent |*uu*> and |*du*> as

$$|uu\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |du\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}. \tag{7.5}$$

Let's see what happens when $\sigma_z \otimes I$ operates on these column vectors. Applying the matrix to |*uu*> results in